LLM & Prompt Engineering

(5th Sem CSE)

Unit-1

1. Identify the type of prompting used in the following instruction: "Translate the sentence into Hindi: 'How are you?'"
2. List any two components of a prompt.
3. What is Zero-shot prompting?
4. What is Temperature in generative AI?
5. Define Prompt Engineering.
6. List any two components of a prompt.
7. What is Zero-shot prompting?
8. Expand LLM and give one example.
9. Who is the Father of Prompt Engineering?
10. List any two Prompt Engineering Techniques.
11. Explain the structure of a prompt with suitable example.
12. Explain the structure of a prompt with suitable example.
13. Differentiate between one-shot and few-shot prompting with examples.
14. Compare between Generative AI & LLMs. (any 6 points)
15. Illustrate the terms a) One-shot prompting b) Role Prompting
16. Discuss in detail what is Prompt Engineering along with its structure and components.
17. Explain Chain-of-thought prompting with a worked-out example.
18. Explain Chain-of-thought prompting with a worked-out example.
19. Generate a prompt JSON output format including AI response and parsing for Product Catalog Entry of e-commerce.

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Unit-2

1. Mention the range value of Top-p parameter of LLM.
2. State any two Pros of Prompt Tuning.
3. State the role of FAISS in AI systems.
4. Which API provides hosted models like GPT and BERT?
5. Identify the prompting pattern used in the following instruction:
   "Let's solve this step-by-step."
6. Which API provides hosted models like GPT and BERT?
7. Describe the use of LangChain in integrating APIs.
8. Give the example to show how temperature parameter control in a language model?
9. Describe the use of LangChain in integrating APIs.
10. Write short notes on:
11. (a)Persona creation in prompting (b) JSON output formatting.
12. Discuss the role of vector databases (FAISS/ChromaDB) in storing and retrieving embeddings.
13. Discuss about the working of Max tokens in LLM.
14. Analyse the steps involved in Persona creation through prompt control.

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Unit-3

1. What is Chroma DB?
2. Tell the role of JSON output formatting.
3. Illustrate HuggingFace Transformers API.
4. With neat diagram, explain how stateful memory is maintained in chatboats. .
5. Name any two vector databases commonly used for storing embeddings.
6. What is the purpose of authentication when using the OpenAI API?
7. Compare FAISS and ChromaDB in terms of scalability, search efficiency, and ease of integration with LLM-based applications. Which would you recommend for a large-scale semantic search system, and why?
8. In a multi-agent LangChain setup, discuss the trade-offs between relying solely on API-based hosted models (like HuggingFace) versus using fine-tuned local models.
9. Design a prompt for building a chatbot that summarizes research papers. Explain its components.
10. With neat diagram, explain how stateful memory is maintained in chatbots.
11. Sketch out the role of FAISS.
12. Describe the Architecture of Long chain in LLM with example.
13. Illustrate HuggingFace Transformers API.

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Unit-4

1. Recall two types of Chatboat.
2. List the core any three components for designing personal AI Assistants.
3. Mention the purpose of Streamlit & Flask in LLM.
4. State one evaluation metric for prompt quality.
5. What is a fallback strategy?
6. What is token usage in the context of AI applications?
7. Why is planning important before starting a mini AI project?
8. Suppose you are developing a personal AI assistant. How would you implement stateful memory to ensure the assistant remembers user preferences across sessions?
9. Compare the challenges of integrating AI applications into web frameworks (like Flask) versus rapid prototyping frameworks (like Streamlit). Which would you choose for a production-grade chatbot, and why?
10. In a multi-agent LangChain setup, discuss the trade-offs between relying solely on API-based hosted models (like HuggingFace) versus using fine-tuned local models.
11. Critically analyze how context retention and UI integration together influence the user experience in an AI-powered translator application.
12. Discuss two cost optimization strategies in generative AI applications.
13. Illustrate the process of latency measurement and performance profiling in AI apps.
14. Design a mini project workflow for building a personal AI assistant using LangChain and APIs. Discuss challenges and solutions.
15. Explain the concept context retention & stateful memory.
16. Design and explain a prompt for building a chatbot for any website that along with its components.

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(5^{th} Sem CSE)

Unit-5

1. How does cost estimation relate to token usage in API-based LLMs?
2. What is a fallback strategy?
3. Suppose your primary model fails to respond in time. How would you implement a fallback strategy using a secondary model?
4. Describe different evaluation metrics for prompt performance with examples.
5. In a multi-model orchestration setup, analyse the trade-offs between using a cheaper but less accurate model versus a more costly but highly accurate one.
6. State one evaluation metric for prompt quality.
7. What is a fallback strategy?
8. Describe different evaluation metrics for prompt performance with examples.
9. Explain with suitable example how multi-model orchestration can reduce cost and improve reliability in generative AI applications.
10. Case Study: You are asked to build an AI-powered translator. Write the detailed prompt design, integration with APIs, evaluation strategy, and fallback mechanism.
11. What is Multi Model Prompt?
12. User feedback is important in Prompt Engineering because…………….
13. Name any 4 Prompt evaluation metrics to enhance productivity.
14. Illustrate the process of latency measurement and performance profiling in AI apps.
15. Generate a prompt JSON output format including AI response and parsing for Product Catalog Entry of e-commerce.
16. Why feedback and peer review is important after final project project demo .Explain it with steps and example. Draw the complete flow of process.